

GDG Management Portal

Modular Monolith Task Allocation Report

Architecture & Distribution Strategy

This document ensures an **exactly equal workload** across all 13 developers by explicitly mapping every single feature (43 Web, 43 App, 24 Backend Endpoints). We are utilizing a strict **Modular Monolith / Feature-Based Architecture**.

Zero-Conflict Rule: Each developer is assigned their own specific feature modules (e.g., modules/tasks/). A developer must place their models, controllers, routes, or UI components strictly inside their assigned module folder. No developer is permitted to write outside their module.

Backend Team (4 Developers) - Node and Express JS

Backend Dev 1: Users & Auth Module

6 Endpoints + 1 DB Model

```
backend/src/dev1_auth_users/  
├─ controllers/  
|   ├── auth.controller.js  
|   └── users.controller.js  
├─ routes/  
|   ├── auth.routes.js  
|   └── users.routes.js  
├─ services/  
|   └── notifs.service.js
```

```
└─ models/
└─ UserModel.js
```

CODE	FEATURE NAME	RESPONSIBILITY
API-01	User Login & Auth	JWT Auth, bcrypt password hashing.
API-02	Get User Profile & Stats	Decode JWT, return stats.
API-03	Team Directory & Search	Text-index search across collections.
API-04	Forgot / Reset Password	OTP/Token generation and email.
API-05	Bulk Member Upload (Admin)	Parse CSV, bulk insert.
API-06	Member Management (Admin)	Modify status, update volatile_data.

Backend Dev 2: Core Business Module (Tasks, Forms, Repo)

6 Endpoints + 5 DB Models

```
backend/src/dev2_tasks_forms/
├─ controllers/
│   ├── tasks.controller.js
│   ├── forms.controller.js
│   ├── leaves.controller.js
│   └─ repo.controller.js
├─ routes/
│   ├── tasks.routes.js
│   ├── forms.routes.js
│   ├── leaves.routes.js
│   └─ repo.routes.js
└─ models/
    ├── TaskModel.js
    ├── FormModel.js
    └─ LeaveModel.js
```

CODE	FEATURE NAME	RESPONSIBILITY
API-08	Fetch Tasks	Pagination and filtering.
API-09	Create & Assign Task	Async trigger notifications.
API-10	Task Workflow	Accept/Submit/Review actions.
API-13	Forms & Submissions	JSON validation, aggregated responses.
API-14	Leave Applications	Approve/Reject leaves.
API-15	Upload & Fetch Assets	Multipart AWS S3 uploads.

Backend Dev 3: Real-Time Chat & Analytics

6 Endpoints (2 REST, 4 WS) + 2 DB Models

```

backend/src/dev3_chat_analytics/
├─ controllers/
│  ├─ chat.controller.js
│  └─ analytics.controller.js
├─ routes/
│  ├─ chat.routes.js
│  └─ analytics.routes.js
├─ sockets/
│  └─ chat.socket.js
└─ models/
   ├─ ChannelModel.js
   └─ ChatModel.js

```

CODE	FEATURE NAME	RESPONSIBILITY
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API-16	Chat Initialization	Fetch channels & history.
WS-01	Socket Connection & Auth	JWT socket auth, subscribe rooms.
WS-02	Join / Leave Channel	Verify access, join room.
WS-03	Send Message	Broadcast msg, process attachments.
WS-04	Typing Indicators	Broadcast typing status.
API-19	Analytics Dashboard	MongoDB aggregation pipelines.

Backend Dev 4: Admin Security, Meetings & Notifications

6 Endpoints (5 REST, 1 WS) + 4 DB Models

```
backend/src/dev4_admin_meetings/
├── controllers/
│   ├── iam.controller.js
│   └── meetings.controller.js
├── routes/
│   ├── iam.routes.js
│   └── meetings.routes.js
├── middlewares/
│   └── rbac.middleware.js
├── sockets/
│   └── notif.socket.js
└── models/
    ├── RoleModel.js
    ├── MeetingModel.js
    └── NotificationModel.js
```

CODE	FEATURE NAME	RESPONSIBILITY
API-07	Manage Roles & Policies	CRUD dynamic roles, RBAC Middleware.

API-
11

Meeting Dashboard

Create and fetch meetings.

API-
12

Attendance & MOM Control

Batch attendance updates.

API-
17

Fetch Notifications

Mark read/unread alerts.

API-
18

Manage Community
Updates

Broadcast internal updates.

WS-05

Real-Time Notifications

Server-to-client push events.

Web Frontend Team (5 Developers) - Next.js and JS

Web Dev 1: Dashboard, Leaves & Repository

9 Features

```
web/src/app/dev1_dashboard_registry/  
├─ dashboard/ (W01-W07)  
│ └─ api/  
│ └─ hooks/  
│ └─ components/  
│   └─ DashboardPage.jsx  
├─ repository/ (W22)  
│ └─ api/  
│ └─ components/  
│   └─ RepoPage.jsx  
└─ leaves/ (W21)  
    └─ api/  
    └─ components/  
      └─ LeavesPage.jsx
```

CODES		FEATURE NAMES
W01	W02	Welcome Message, Member/Task Stats, Today's Meeting
W03		
W04	W05	My Tasks, Quick Action Buttons, Community Updates Graph
W06		
W07	W21	Global Search Bar, Leave Application Management, Essential Documents
W22		

Web Dev 2: Tasks, Calendar & Registry Tables

9 Features

```
web/src/app/dev2_tasks_calendar/  
├─ tasks/ (W08-W13)  
│ └─ api/  
│ └─ hooks/  
│ └─ components/  
│   └─ TasksPage.jsx  
├─ registry/ (W19-W20)  
│ └─ api/  
│ └─ components/  
│   └─ RegistryPage.jsx  
└─ docs/ (W32)  
    └─ api/  
    └─ components/  
      └─ DocsPage.jsx
```

CODES		FEATURE NAMES
W08	W09	Calendar View, Tasks List View, Add Task Button
W10		
W11	W12	Task Filters, Task Workflow & Table, Task Activity Log
W13		

Web Dev 3: Meetings & Dynamic Forms

9 Features

```
web/src/app/dev3_meetings_forms/  
├─ meetings/ (W14-W18)  
│ └─ api/  
│ └─ hooks/  
│ └─ components/  
│ └─ MeetingsPage.jsx  
└─ forms/ (W23-W26)  
    └─ api/  
    └─ hooks/  
    └─ components/  
    └─ FormsPage.jsx
```

CODES

FEATURE NAMES

W14 W15

W16

Meeting Dashboards, Create Meeting, Meeting Filters

W17 W18

W23

Summary & Attendance, Integrated Meeting Link, Active Form Cards

W24 W25

W26

Form Fillup Fields, Form Instructions, Form Responses Dashboard

Web Dev 4: Profile, Auth & User Settings

9 Features

```
web/src/app/dev4_profile_auth/
```

```
└─ auth/ (W31, W36)
  └─ api/
  └─ hooks/
  └─ components/
    └─ LoginPage.jsx
└─ profile/ (W27-W30)
  └─ api/
  └─ components/
    └─ ProfilePage.jsx
└─ communications_misc/ (W33-W35)
  └─ api/
  └─ components/
    └─ MiscPage.jsx
```

CODES

FEATURE NAMES

W27 W28 W29

User Profile, Sidebar Options, Dark / Light Mode

W30 W31 W33

Developer Mode, Login & Auth, Terms and Conditions

W34 W35 W36

Notification Center (Bell), Team Directory, Forgot Password Flow

Web Dev 5: Admin IAM & Real-Time Chat

7 Features (High Complexity)

```
web/src/app/dev5_admin_chat/
└─ admin/ (W37-W41)
  └─ api/
  └─ hooks/
  └─ components/
    └─ AdminDash.jsx
└─ chat/ (W42-W43)
  └─ api/
  └─ hooks/
  └─ components/
    └─ ChatApp.jsx
```


CODES	FEATURE NAMES
W37 W38 W39	Admin Dashboard, Member Management, Application Approvals
W40 W41	Dynamic Roles & IAM UI, Bulk Member Upload (CSV)
W42 W43	Core Chat (WebSockets), Advanced Chat (Files & Mentions)

Mobile App Team (4 Developers) - Flutter (Dart)

App Dev 1: Dashboard, Registry & Leaves

11 Features

```
app/lib/features/dev1_dashboard_registry/  
├─ screens/  
│   ├── DashboardScreen.dart  
│   ├── WorksTable.dart  
│   ├── MeetingTable.dart  
│   ├── LeavesScreen.dart  
│   └── RepoScreen.dart  
└─ components/  
    ├── UpdatesGraph.dart  
    └── GlobalSearch.dart
```

CODES	FEATURE NAMES
A01 A02 A03	Welcome Message, Member/Task Stats, Today's Meeting
A04 A05 A06	My Tasks, Quick Action Buttons, Updates Graph
A07 A19 A20	Global Search, Tables of Works, Tables of Meeting

A21

A22

Leave Application, Essential Documents

App Dev 2: Tasks, Forms & Notifications

11 Features

```
app/lib/features/dev2_tasks_forms/  
├─ screens/  
│   ├─ TasksScreen.dart  
│   ├─ CalendarScreen.dart  
│   ├─ FormsScreen.dart  
│   └─ FormResponses.dart  
└─ components/  
    ├─ ActivityLog.dart  
    ├─ DynamicFields.dart  
    └─ NotifBell.dart
```

CODES

FEATURE NAMES

A08

A09

A10

Calendar View, Tasks List View, Add Task Button

A11

A12

A13

Task Filters, Task Workflow, Task Activity Log

A23

A24

A25

Active Forms, Form Fillup Fields, Form Instructions

A26

A34

Form Responses Dashboard, Notification Center (Bell)

App Dev 3: Meetings, Profile & Auth

11 Features

```
app/lib/features/dev3_meetings_profile/  
├─ screens/
```

```
| |─ MeetingsScreen.dart
| |─ MeetDetails.dart
| |─ ProfileScreen.dart
| |─ LoginScreen.dart
| |─ DirectoryScreen.dart
| └─ components/
|   └─ SidebarDrawer.dart
|   └─ ThemeToggle.dart
```

CODES

FEATURE NAMES

A14 A15 A16

Meeting Dashboards, Create Meeting, Meeting Filters

A17 A18 A27

Summary & Attendance, Integrated Meeting, User Profile

A28 A29 A31

Sidebar Options, Dark / Light Mode, Login & Auth

A35 A36

Team Directory, Forgot Password Flow

App Dev 4: Admin IAM, Real-Time Chat & Misc

10 Features (High Complexity)

```
app/lib/features/dev4_admin_chat/
└─ screens/
  | └─ AdminScreen.dart
  | └─ IAMConfigScreen.dart
  | └─ ChatScreen.dart
  | └─ DocsScreen.dart
  |   └─ TermsScreen.dart
  └─ components/
    └─ DevModeToggle.dart
```

CODES

FEATURE NAMES

A37

A38

A39

Admin Dashboard, Member Management, Application Approvals

A40

A41

A42

IAM UI, Bulk Member Upload, Core Chat

A43

A30

A32

Advanced Chat (Files/Mentions), Developer Mode,
Documentation

A33

Terms and Conditions

Detailed Folder Structures

This is the complete bird's-eye view of the project repositories. Notice how every single platform strictly isolates its code into feature modules to prevent any Git conflicts.

1. Backend Repository (Node.js/Express)

```
backend/src/
├── dev1_auth_users/ <-- Backend Dev 1 works ONLY here
│   ├── controllers/
│   │   ├── auth.controller.js (API-01, API-04)
│   │   └── users.controller.js (API-02, API-03, API-05, API-06)
│   ├── routes/
│   │   ├── auth.routes.js
│   │   └── users.routes.js
│   ├── services/
│   │   └── notifs.service.js (API-17, API-18)
│   ├── models/
│   └── UserModel.js
├── dev2_core_business/ <-- Backend Dev 2 works ONLY here
│   ├── controllers/
│   │   ├── tasks.controller.js (API-08, API-09, API-10)
│   │   └── forms.controller.js (API-13)
```

```

├── leaves.controller.js (API-14)
├── repo.controller.js (API-15)
├── routes/
│   ├── tasks.routes.js
│   ├── forms.routes.js
│   ├── leaves.routes.js
│   └── repo.routes.js
├── models/
│   ├── TaskModel.js
│   ├── FormModel.js
│   ├── FormSubmissionModel.js
│   ├── LeaveApplicationModel.js
│   └── RepositoryModel.js
├── dev3_realtime_chat/ <-- Backend Dev 3 works ONLY here
│   ├── sockets/
│   │   └── chat.socket.js (WS-01, WS-02, WS-03, WS-04)
│   ├── controllers/
│   │   ├── chat.controller.js (API-16)
│   │   └── stats.controller.js (API-19)
│   ├── routes/
│   │   ├── chat.routes.js
│   │   └── analytics.routes.js
│   └── models/
│       ├── ChannelModel.js
│       └── ChatModel.js
├── dev4_admin_meetings/ <-- Backend Dev 4 works ONLY here
├── controllers/
│   ├── iam.controller.js (API-07)
│   └── meet.controller.js (API-11, API-12)
├── sockets/
│   └── alerts.socket.js (WS-05)
├── routes/
│   ├── iam.routes.js
│   └── meetings.routes.js
├── models/
│   ├── RoleModel.js
│   ├── PolicyModel.js
│   ├── MeetingModel.js
│   ├── AttendanceModel.js
│   ├── NotificationModel.js
│   └── UpdatesModel.js

```

2. Web Frontend Repository (React/Next.js)

```
web/src/
├── shared/ <-- Global Shared Resources
│   ├── components/
│   ├── hooks/
│   ├── api/
│   │   ├── axios.js
│   │   └── apiClient.js
│   ├── utils/
│   ├── constants/
│   └── assets/
├── features/
├── dev1_dashboard_repo/ <-- Web Dev 1 works ONLY here
│   ├── dashboard/
│   │   ├── api/
│   │   ├── hooks/
│   │   ├── components/
│   │   │   ├── UpdatesGraph.jsx (W06)
│   │   │   └── GlobalSearch.jsx (W07)
│   │   ├── services/
│   │   ├── utils/
│   │   └── DashboardPage.jsx (W01-W05)
│   ├── leaves/
│   │   ├── api/
│   │   ├── components/
│   │   └── LeavesPage.jsx (W21)
│   ├── repository/
│   │   ├── api/
│   │   ├── components/
│   │   └── RepositoryPage.jsx (W22)
├── dev2_tasks_registry/ <-- Web Dev 2 works ONLY here
│   ├── tasks/
│   │   ├── api/
│   │   ├── hooks/
│   │   ├── components/
│   │   │   └── ActivityLog.jsx (W13)
│   │   ├── services/
│   │   └── TasksPage.jsx (W09-W12)
│   ├── calendar/
│   │   ├── api/
│   │   ├── components/
│   │   └── CalendarPage.jsx (W08)
│   ├── registry/
│   │   ├── api/
│   │   ├── components/
│   │   ├── WorksTablePage.jsx (W19)
│   │   ├── MeetingTable.jsx (W20)
│   │   └── DocsPage.jsx (W32)
```

```
|— dev3_meetings_forms/ <-- Web Dev 3 works ONLY here
|   |— meetings/
|   |   |— api/
|   |   |— hooks/
|   |   |— components/
|   |   |— MeetingsPage.jsx (W14-W16, W18)
|   |   |— MeetingDetails.jsx (W17)
|   |— forms/
|   |   |— api/
|   |   |— hooks/
|   |   |— components/
|   |   |   |— DynamicFields.jsx (W24)
|   |— ActiveForms.jsx (W23, W25)
|   |— FormResponses.jsx (W26)
|
|— dev4_auth_profile/ <-- Web Dev 4 works ONLY here
|   |— auth/
|   |   |— api/
|   |   |— hooks/
|   |   |— components/
|   |   |— LoginPage.jsx (W31, W36)
|   |   |— TermsPage.jsx (W33)
|   |— profile/
|   |   |— api/
|   |   |— components/
|   |   |— ProfilePage.jsx (W27)
|   |   |— DirectoryPage.jsx (W35)
|   |— layout/
|   |   |— components/
|   |   |   |— Sidebar.jsx (W28)
|   |   |   |— ThemeToggle.jsx (W29)
|   |   |   |— DevModeToggle.jsx (W30)
|   |   |   |— NotifBell.jsx (W34)
|
|— dev5_admin_chat/ <-- Web Dev 5 works ONLY here
|   |— admin/
|   |   |— api/
|   |   |— hooks/
|   |   |— components/
|   |   |— AdminDash.jsx (W37-W39, W41)
|   |   |— IAMConfig.jsx (W40)
|   |— chat/
|   |   |— api/
|   |   |— hooks/
|   |   |— components/
|   |   |   |— ChatSidebar.jsx
|   |   |   |— ChatWindow.jsx
|   |— ChatApp.jsx (W42, W43)
```

3. App Frontend Repository (Flutter)

```
app/lib/features/
├── dev1_dashboard_registry/ <-- App Dev 1 works ONLY here
│   ├── screens/
│   │   ├── DashboardScreen.dart (A01, A02, A03, A04, A05)
│   │   ├── WorksTable.dart (A19)
│   │   ├── MeetingTable.dart (A20)
│   │   ├── LeavesScreen.dart (A21)
│   │   └── RepoScreen.dart (A22)
│   └── components/
│       ├── UpdatesGraph.dart (A06)
│       └── GlobalSearch.dart (A07)
├── dev2_tasks_forms/ <-- App Dev 2 works ONLY here
│   ├── screens/
│   │   ├── TasksScreen.dart (A09, A10, A11, A12)
│   │   ├── CalendarScreen.dart (A08)
│   │   ├── FormsScreen.dart (A23, A25)
│   │   └── FormResponses.dart (A26)
│   └── components/
│       ├── ActivityLog.dart (A13)
│       ├── DynamicFields.dart (A24)
│       └── NotifBell.dart (A34)
├── dev3_meetings_profile/ <-- App Dev 3 works ONLY here
│   ├── screens/
│   │   ├── MeetingsScreen.dart (A14, A15, A16, A18)
│   │   ├── MeetDetails.dart (A17)
│   │   ├── ProfileScreen.dart (A27)
│   │   ├── LoginScreen.dart (A31, A36)
│   │   └── DirectoryScreen.dart (A35)
│   └── components/
│       ├── SidebarDrawer.dart (A28)
│       └── ThemeToggle.dart (A29)
├── dev4_admin_chat/ <-- App Dev 4 works ONLY here
│   ├── screens/
│   │   ├── AdminScreen.dart (A37, A38, A39, A41)
│   │   ├── IAMConfigScreen.dart (A40)
│   │   ├── ChatScreen.dart (A42, A43)
│   │   ├── DocsScreen.dart (A32)
│   │   └── TermsScreen.dart (A33)
│   └── components/
│       └── DevModeToggle.dart (A30)
```


